

# Introduction

This documentation explains the structure and design of a music database. The database stores information about songs, artists, albums, and playlists. It helps users manage and search music easily. This guide describes the tables, relationships, and how data is organized. It is intended for anyone who wants to understand or use the database.



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## Get to know About Data

The music database contains all the information about artists, albums, tracks, and customers. It shows how music is created, sold, and listened to.

- Artists are the people or bands who make the music.
- Albums are collections of songs made by artists.
- Tracks are the individual songs that people can listen to or buy.
- Customers are the people who buy or stream the music.
- Invoices and InvoiceLines show which tracks were purchased, when, and by whom.

This data helps us understand music trends, popular artists, top-selling albums, and customer behavior. It also allows us to calculate key metrics, like which songs earn the most or which customers are the most active.

In short, the database tells the full story of music creation, consumption, and sales, all connected through simple relationships between artists, albums, tracks, and customers

## ➤ Defining The columns and Data Types

- **Purpose:** Stores all information about the people who buy or interact with the music.

| Column Name | Description | Data Type |

|-----|-----|-----|

- **Table Name: customers**

| CustomerId | Unique ID for each customer | int |

| FirstName | Customer's first name | varchar |

| LastName | Customer's last name | varchar |

| Email | Customer's email address | varchar |

| Phone | Customer's phone number | varchar |

| Address | Customer's street address | varchar |

| City | Customer's city | varchar |

| State | Customer's state | varchar |

| PostalCode | Customer's postal/ZIP code | varchar |

| Country | Customer's country | varchar |

| Fax | Customer's fax number | varchar |

### Table Name: Album

- **Purpose:** Stores all albums released by artists.

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| AlbumId | Unique ID for each album | int |

| Title | Name of the album | varchar |

| ArtistId | ID of the artist who released the album | int |

### Table Name: Artist

- **Purpose:** Stores information about all artists (singers or bands).

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| ArtistId | Unique ID for each artist | int |

| Name | Name of the artist | varchar |

## Table Name: Employee

- **Purpose:** Stores all information about employees who manage sales or work with customers.

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| EmployeeId | Unique ID for each employee | int |

| FirstName | Employee's first name | varchar |

| LastName | Employee's last name | varchar |

| Title | Job title of the employee | varchar |

| BirthDate | Employee's date of birth | date |

| ReportsTo | ID of the manager/employee they report to | int |

| HireDate | Date the employee was hired | date |

| Address | Employee's street address | varchar |

| City | Employee's city | varchar |

| State | Employee's state | varchar |

| Country | Employee's country | varchar |

| PostalCode | Postal or ZIP code | varchar |

| Phone | Employee's phone number | varchar |

| Fax | Employee's fax number | varchar |

| Email | Employee's email address | varchar |

### Table Name: Genre

- **Purpose:** Stores all music genres available in the database.

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| GenreId | Unique ID for each genre | int |

| Name | Name of the genre (e.g., Rock, Pop, Jazz) | varchar |

### Table Name: Invoice

- **Purpose:** Stores all purchase transactions made by customers.

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| InvoiceId | Unique ID for each invoice | int |

| InvoiceDate | Date of the invoice | date |

| BillingAddress | Customer billing address | varchar |

| BillingCity | Customer billing city | varchar |

| BillingState | Customer billing state | varchar |

| PostalCode | Postal or ZIP code | varchar |

| Total | Total amount of the invoice | decimal |

| CustomerId | ID of the customer who made the purchase | int |

### Table Name: InvoiceLine

- **Purpose:** Stores details of each track included in an invoice (which songs were bought and their quantity).

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| InvoiceLineId | Unique ID for each line in the invoice | int |

| InvoiceId | ID of the invoice this line belongs to | int |

| TrackId | ID of the track being purchased | int |

| UnitPrice | Price of the track | decimal |

| Quantity | Number of tracks purchased | int |

### Table Name: MediaType

- **Purpose:** Stores the types of media for tracks (e.g., MP3, WAV, AAC).

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| MediaTypeId | Unique ID for each media type | int |

| Name | Name of the media type | varchar |

### Table Name: Playlist

- **Purpose:** Stores all playlists created in the database. Playlists are collections of tracks grouped by theme, mood, or user preference.

- **Columns / Data Types:**

| Column Name | Description | Data Type |
|-------------|-------------|-----------|
|-------------|-------------|-----------|

|       |       |       |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

|            |                             |     |
|------------|-----------------------------|-----|
| PlaylistId | Unique ID for each playlist | int |
|------------|-----------------------------|-----|

|      |                      |         |
|------|----------------------|---------|
| Name | Name of the playlist | varchar |
|------|----------------------|---------|

### Table Name: PlaylistTrack

- **Purpose:** Links tracks to playlists. Shows which tracks are in each playlist.

- **Columns / Data Types:**

| Column Name | Description | Data Type |
|-------------|-------------|-----------|
|-------------|-------------|-----------|

|       |       |       |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

|            |                    |     |
|------------|--------------------|-----|
| PlaylistId | ID of the playlist | int |
|------------|--------------------|-----|

|         |                                          |     |
|---------|------------------------------------------|-----|
| TrackId | ID of the track included in the playlist | int |
|---------|------------------------------------------|-----|

**Note:** This is a many-to-many relationship table because a playlist can have many tracks, and a track can be in many playlists.

## Table Name: Track

- **Purpose:** Stores all songs/tracks available in the database.

- **Columns / Data Types:**

| Column Name | Description | Data Type |

|-----|-----|-----|

| TrackId | Unique ID for each track | int |

| Name | Name of the track | varchar |

| AlbumId | ID of the album this track belongs to | int |

| MediaTypeId | ID of the media type (MP3, WAV, etc.) | int |

| GenreId | ID of the music genre | int |

| Composer | Name of the composer | varchar |

| Milliseconds | Duration of the track in milliseconds | int |

| Bytes | File size in bytes | int |

| UnitPrice | Price of the track | decimal |

